

Energy-10 Developments



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Q. It's rumored that the release of a major new version of *Energy-10* is imminent, though I heard that development of *Energy-10* had stopped several years ago. What would you say distinguishes *Energy-10* from the many alternatives for building modeling? What exactly are the new features in this next release, and what's in store for the future of *Energy-10*?

—Modeler in Moline

A. *Energy-10* is a software package designed to make it easy to look at energy efficiency strategies for buildings early in the design process—at the point at which it might actually be possible to make dramatic improvements in a design while working with the architecture. *Energy-10* encourages using an integrated design approach. “Under the hood,” *Energy-10* uses an hourly simulation of the building envelope and HVAC systems.

Energy-10 is a unique tool that can rapidly tell you the range of possible energy savings starting from minimal user input, and can quickly identify and rank the most promising directions to explore. It shines in the context of the predesign/schematic design phase of the process. There are indeed a number of alternatives now for building energy modeling. J. Douglas Balcomb began developing *Energy-10* in 1992 at the National Renewable Energy Laboratory (NREL). At that time, options were much more limited and the computer simulation tools available were more for the research community. If integrated design was to be accomplished, Bal-

comb saw the need for a more accessible, architect-friendly tool.

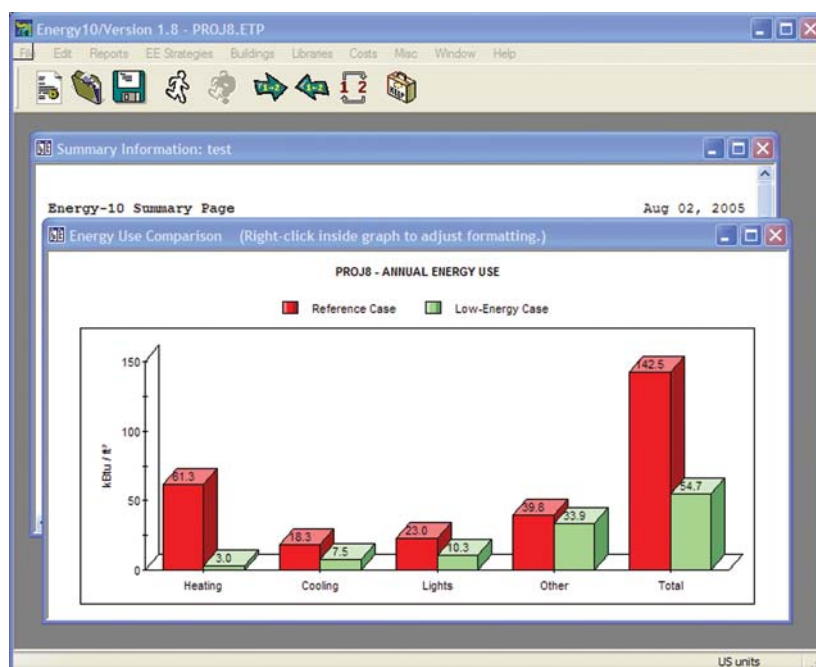
Although development funding provided by DOE has been reduced, work has continued. Some work was under way several years ago to introduce a visual building editor and to change the underlying structure of *Energy-10* for more flexible definition of multizone buildings. It turned out that these changes could not be made without a complete rewrite of the software, for which funds were not available. Since then, we've focused on more incremental improvements.

One key capability in *Energy-10* is the rapid creation of a reference case and a low-energy case. You can create these models literally in seconds with *Energy-10*, starting from four inputs—location, building footprint, building use profile, and HVAC system type. The Autobuild process, as it is called, creates a simple rectangular geometry as a reference case representing standard practice. Then a number of predefined energy efficiency strategies (EES) are automati-

cally applied, including envelope, daylighting, lighting, and HVAC systems measures. These two cases tell you what ballpark you are in—what general magnitude of savings may be achieved. *Energy-10* can then be used to rank all the EES to suggest the most profitable design directions with respect to energy efficiency. Full lifecycle costing is embedded in the tool, along with many useful graphs.

I think *Energy-10* embraces the philosophy that it's important early in the design process not to get lost in the energy-modeling woods—to explore strategies and get a sense of the more promising directions for a given design based on a manageable, simple model. Many of the most important energy efficiency insights can be extracted from a simple model, saving the more detailed geometry and/or multizone thermal models for specific design questions that arise during design development. For many projects, *Energy-10* is all you need.

Energy-10 Version 1.7 recently hit the streets. This version provides several new



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HVAC system options and a new way of handling windows that uses National Fenestration Rating Council performance parameters. Version 1.8 is currently in beta test and should be distributed by the Sustainable Buildings Industry Council (SBIC) within several months. (For Version 1.7 and Version 1.8 when it becomes available, go to www.sbicouncil.org.) SBIC recently obtained the distribution license for *Energy-10* from NREL and SBIC will provide workshops and an online user forum.

Version 1.8 will be a significant revision of *Energy-10*, with additions funded by DOE through the NREL Solar Program. The key new features of *Energy-10* include integrated modeling of PV systems (grid-tied systems only in this current release) and solar thermal domestic hot water modeling. My colleagues on the development team and I believe that these modeling capabilities are useful for exploring, among other things, net-zero energy use strategies for residences and

small commercial buildings. What may be unique to *Energy-10* is the automatic EES process for including PV in your design. If you apply building-integrated PV in *Energy-10*, both the PV system model and the changes to the building envelope are defined. Thus, the simulation can show the combined effect of modifications to the envelope and electric load offset by the PV system. (The underlying simulation of PV uses the Sandia National Laboratory module performance model by calling an embedded version of the *EnergyPlus* simulation engine.) A simple hourly solar domestic hot water (DHW) model can be run to evaluate the potential to reduce hot water loads on electric and gas water heaters.

In Version 1.8, we also introduce a new library with predefined constructions intended to support Leadership in Energy and Environmental Design (LEED)-type performance rating studies based on ASHRAE 90.1-2004, Appendix G. Finally, the look of *Energy-10* has

been updated, along with the extensive online help system.

We're excited about the new distribution arrangement with SBIC and look forward to ongoing improvements in the software that will be driven by *Energy-10* users.



*The Midwest Research Institute (MRI), management and operating contractor for NREL, developed **Energy-10** Version 1.7 and holds certain intellectual property rights, including trademark and copyright rights related to **Energy-10** Version 1.7 software that it has licensed to SBIC for the purpose of commercialization.*

For more information:

Go to www.sbicouncil.org.